

Creatine as a food supplement for the general population

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ABSTRACT

Creatine is a non-proteinogenic amino acid available from various animal-based foods or synthesized endogenously in the human body. A number of recent population-based studies demonstrate a lower-than-expected dietary creatine intake across age- and gender-specific cohorts, with low creatine consumption accompanied by various health risks. Those studies suggest that the general public may benefit from creatine to prevent and manage various health conditions or to ensure advanced growth. Favorable safety and promising impact of supplemental creatine on human well-being and functioning emanated from plenty of small-sampled interventional studies perhaps suggest a need for recommending creatine to the general public. In this opinion paper, I have outlined the possible rationales for endorsing supplemental creatine ubiquitously, and discussed opportunities and challenges for population-wide creatine use.

1. Introduction

Creatine is arguably one of the most popular dietary supplements worldwide, with the global creatine market estimated to be \$520 million by 2024 (360 Research Report, 2019). Identified as a natural component of skeletal muscle in 1832 by French chemist Michel Eugène Chevreul (Wallimann, 2007), creatine has started to be investigated as a food additive as early as 1912 (Folin & Denis, 1912). The research on creatine skyrocketed in the early 1990 s, and almost 10,000 scientific reports up to this point have evaluated its effects across various domains of nutrition and biomedicine. Beyond the shadow of a doubt, enhancing exercise performance remains a central field of creatine supplementation (Kreider et al., 2017). However, groundbreaking studies published in the past 20 years confirmed its beneficial effects outside of the athletic arena, including neuromuscular and cardiometabolic diseases, post-traumatic rehabilitation, and aging (Kreider & Stout, 2021). A promising impact of supplemental creatine on human well-being and functioning that emanated from small-sized interventional trials perhaps opens a window of opportunity for recommending creatine to the general public. In this opinion paper, I have outlined possible rationales for endorsing supplemental creatine ubiquitously and discussed opportunities and challenges for population-wide creatine use.

2. Creatine turnover: Loss vs. Replacement

Creatine ($C_4H_9N_3O_2$) is a non-proteinogenic amino acid that plays a crucial role in cellular bioenergetics. This guanidino compound (also known as methyl-guanidinoacetic acid) is found in vertebrates, where it facilitates the recycling of high-energy phosphates that provide energy to cellular processes. Consequently, creatine is abundant in tissues with high-level energy demands, with approximately 95% of the human body's creatine stores being found in the skeletal muscle, with the remaining 5% being distributed in the brain, liver, kidney, and testes (Persky & Brazeau, 2001). An average person requires ~ 2 g of creatine per day, with about half of that amount getting synthesized endogenously in the liver and kidney, and another half obtained from an omnivorous diet (Brosnan & Brosnan, 2007) (Fig. 1). Conversely, creatine spontaneously degrades into creatinine and is excreted from the body by the kidney, with the rate of loss is estimated to be around 1.7% of the total body pool per day (Wyss & Kaddurah-Daouk, 2000).

A balance between continual creatine utilization and substitution appears to be essential in maintaining normal cellular energy flow (Kazak & Cohen, 2020). This equilibrium could be often threatened by a deficient creatine supply, either due to a compromised synthesis of creatine inside the body or diets poor in creatine-containing foods, leading to various conditions with impaired energy metabolism. For

Abbreviations: ADP, adenosine diphosphate; ATP, adenosine triphosphate; CAERS, Center for Food Safety and Applied Nutrition Adverse Event Reporting System; CDS, creatine deficiency syndromes; FDA, Food and Drug Administration; GAA, guanidinoacetic acid; NHANES, National Health and Nutrition Examination Survey; SAH, S-Adenosyl-L-Homocysteine; SAM, S-Adenosyl-L-methionine.

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instance, creatine deficiency syndromes (CDS), a group of inborn errors of creatine biosynthesis and transport, are characterized by a substantial lack of creatine in conjunction with a severe neurological phenotype (Schulze, 2013). Creatine turnover is compromised in vegans and vegetarians who experience low creatine availability owing to the insufficient intake of creatine from food sources despite the augmented internal synthesis (Balestrino & Adriano, 2019). Creatine shortfall is also seen in a plethora of clinical disorders, including liver disease (Verma et al., 2008), autism spectrum disorder (Goji et al., 2017), multiple sclerosis (Ostojic, 2020a), and post-viral fatigue syndrome (Ostojic, 2021a), to name but a few. Interestingly, most of the above conditions positively respond to creatine supplementation by means of recovering creatine levels and its homeostatic turnover, and improving clinical symptoms, functioning, and quality of life.

3. Creatine consumption in the general population

Creatine is available from a variety of animal-based foods – from human breast milk and infant formulas to meat, poultry, and fish, to

dietary supplements containing artificially prepared and purified creatine, typically as creatine monohydrate. The amount of creatine across natural food sources varies from 0.2 g/kg in breast milk to 11.0 g/kg in raw and dried herring fillet; plant-based foods do not contain any creatine (Wu, 2020). Creatine provision from food is expected to contribute ~ 1 g to the daily net yield, and the mean intake may vary as a function of gender and age (Brosnan & Brosnan, 2007). However, a number of recent population-based studies demonstrate a lower-than-expected dietary creatine intake across various age- and gender-specific cohorts. Bakian, Huber, Scholl, Renshaw, and Kondo (2020) were arguably the first who reported a possible creatine shortage from food in the U.S. population, with the average intake almost 50% lower than anticipated. The authors analyzed 22,692 participants ≥ 20 years of age using data from the 2005 to 2012 National Health and Nutrition Examination Survey (NHANES), and calculated the mean daily creatine intake of 0.54 g across all participants. Another trial evaluated dietary creatine intake in 9254 responders aged 0–85 years who participated in the NHANES 2017–2018 round (Ostojic, 2021b). The average creatine intake reported was 1.38 g/day, yet as much as 42.8% of participants

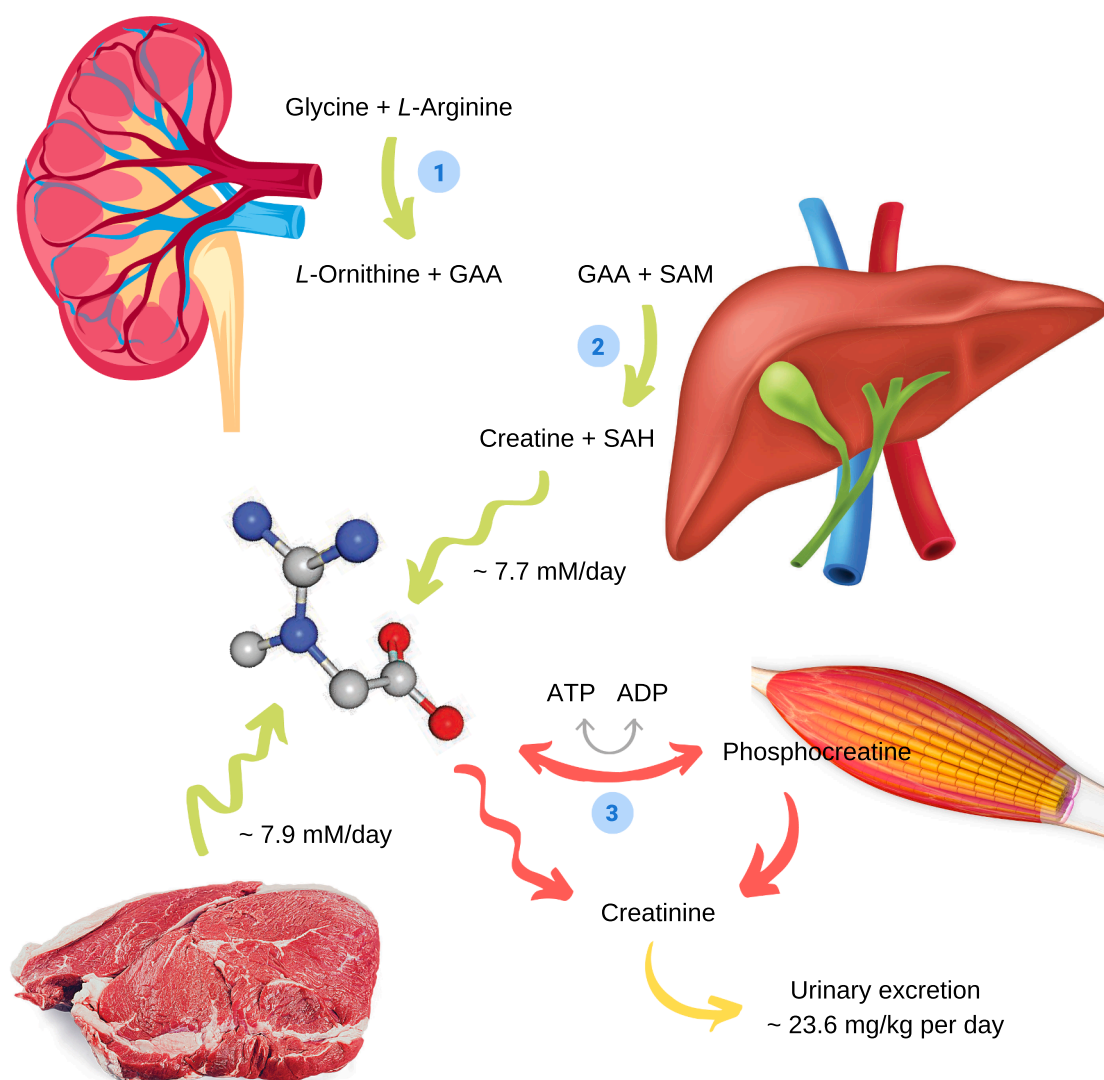


Fig. 1. Metabolism and daily turnover of creatine. Numeric values for creatine replacement via diet and endogenous synthesis, and creatinine excretion represent average amounts for men aged 20–39 years and 18–29 years, respectively. Green arrows indicate creatine provision through a meat-based diet or endogenous synthesis, while red arrows show creatine utilization and non-enzymatic transformation. Controlling enzymes are identified with blue circles as (1) L-arginine-glycine amidinotransferase, EC 2.1.4.1, (2) guanidinoacetate N-methyltransferase, EC 2.1.1.2, and (3) creatine kinase, EC 2.7.3.2. Abbreviations: GAA – guanidinoacetic acid; SAM – S-Adenosyl-L-methionine; SAH – S-Adenosyl-L-Homocysteine; ATP – adenosine triphosphate; ADP – adenosine diphosphate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

were found to have dietary creatine intake below expected levels of one gram per day. Daily intake of creatine across the sample of 4291 children and adolescents aged 2–19 years was 1.07 g (Korovljev, Stajer, & Ostojic, 2021); at the same time, about 60% of young participants consumed creatine below a threshold level of 1 g per day, and 17% consumed zero creatine. The above studies suggest a relatively extensive creatine deficiency at the populational level, with perhaps every second individual consuming too little creatine from food.

A possible explanation contributing to inadequate creatine consumption might include lower meat consumption in the general public. Two-thirds of Americans reported reducing meat consumption from 2012 to 2015, with a reduction of red meat (the predominant dietary source of creatine) being most frequent (Neff et al., 2018). Whether low dietary intake of creatine is fully and easily compensated by enhanced endogenous synthesis remains unknown at this moment. Interestingly, preliminary evidence shows a strong relationship between creatine intake and health risks or growth indicators at the populational level. For instance, depression prevalence was 42% higher among U.S. adults in the lowest quartile of dietary creatine consumption (0–0.26 g/day) compared to participants in the highest quartile of creatine intake (0.70–3.16 g/day) (Bakian et al., 2020). After controlling for demographic and lifestyle variables, the risk of screening positive for depression remained 31% lower among adults in the highest, compared to the lowest quartile of creatine consumption. Our group has found that the shortage of dietary creatine is associated with an increased risk of medical conditions in men and women aged 65 years and over, with elderly who consume <1 g of creatine daily having 2.62 times higher risk of angina pectoris and 2.59 times higher risk of liver conditions, compared to older counterparts who consume >1 g of creatine per day after controlling for demographic and nutritional variables (Ostojic et al., unpublished data). In addition, the daily intake of creatine from a regular diet in shorter children and adolescents was lower than in taller peers aged 2–19 years (Korovljev et al., 2021); each additional 0.1 g of creatine consumed per day increases height by 0.60 cm (simple model) or 0.30 cm (adjusted model). Those studies help to put forward the idea that the general public may benefit from increased creatine intake from food sources in the prevention and management of various health conditions or to ensure advanced growth.

4. Clinical relevance of creatine supplementation

Creatine supplementation may offer various benefits across the range of life-stage groups, medical pathologies, and sport disciplines. Here, I bring into focus possible advantages of creatine use in the general public and medicine while extensive evidence about its application in sports and exercise nutrition is available elsewhere (Kreider et al., 2017); co-administration studies that provided creatine along with exercise regimen or other therapeutics were not included in this summary. Still, the health benefits of creatine are often augmented when it is combined with resistance training in clinical and non-clinical populations (Burke et al., 2003; Candow, Forbes, Kirk, & Duque, 2021; Forbes, Candow, Ferreira, & Souza-Junior, 2021).

The era of creatine use in contemporary medicine presumably started back in 1985, when the researchers from the University of Helsinki Children's Hospital demonstrated favorable effects of 5-year creatine supplementation in 13 patients with gyrate atrophy of the choroid and retina, a rare autosomal recessive metabolic disease (Vannas-Sulonen, Sipilä, Vannas, Simell, & Rapola, 1985). Supplemental creatine (0.25 to 0.50 g three times daily) was ineffective in slowing down the natural progress of the disease for ocular features. Nevertheless, muscle atrophy and tubular aggregates, which characteristically appear early in patients with gyrate atrophy, either disappeared or diminished rapidly during the treatment and remained absent in repeated biopsies during creatine supplementation. These initial findings encouraged health professionals, slowly but surely, to experiment with supplemental creatine across various domains, from mild metabolic ailments and aging-associated

conditions to neuromuscular disorders, cardiometabolic diseases, and cancer. For instance, supplemental creatine (5 g four times a day for five days and then twice a day for 51 days) reduces total plasma cholesterol, very-low-density lipoproteins, and triacylglycerols levels in mildly hyperlipidemic men and women aged 32 to 70 years (Earnest, Almada, & Mitchell, 1996), implying that creatine may modulate lipid metabolism in certain individuals. An additional benefit of creatine on cardiometabolic profiles in the blood was shown in another interventional trial, where 5 g/day of creatine for three weeks reduces plasma levels of homocysteine, a well-known risk factor for heart disease, in strict vegan subjects (Van Bavel, de Moraes, & Tibirica, 2019); this was accompanied by an enhanced skin capillary perfusion. A vasoprotective effect of creatine has been corroborated in another trial with healthy young adults, where one-week creatine supplementation (20 g/day) improves systemic endothelial-dependent microvascular reactivity and increases skin capillary density and recruitment (Moraes, Van Bavel, Moraes & Tibirica, 2014). Furthermore, creatine improved prolonged cognitive performance in healthy participants (Van Cutsem et al., 2020), alleviated decrements in skill execution in sleep-deprived volunteers (Cook, Crewther, Kilduff, & Gaviglio, 2011), improved cognitive functioning and memory in vegetarians (Benton & Donohoe, 2011) and non-vegetarians (Ling, Kritikos, & Tiplady, 2009), and delayed neuromuscular fatigue, improved work capacity and cognition in elderly (Gotshalk et al., 2002; Stout et al., 2007; Morris, Mielcarz, Harris, Swain, & Howard, 2007; Gotshalk et al., 2008).

In the clinical environment, supplemental creatine improved symptomatic and endoscopic indices of Chron's ileitis (Roy & Lee, 2016), increased muscle mass in rheumatoid arthritis (Wilkinson et al., 2016), improved muscle function in fibromyalgia patients (Alves et al., 2013), patients with congestive heart failure (Kuethe, Krack, Richartz, & Figulla, 2006) and Duchenne muscular dystrophy (Felber et al., 2000; Louis et al., 2003; Banerjee et al., 2010), improved developmental outcomes in CDS (Ganesan, Johnson, Connelly, Eckhardt, & Surtees, 1997; Leuzzi et al., 2000; Schulze, Ebinger, Rating, & Mayatepek, 2001; Battini et al., 2006; Ndika et al., 2012) and mitochondrial encephalopathy (Barisic et al., 2002), made better outcome measures in resistant depression (Roitman, Green, Osher, Karni, & Levine, 2007), and showed beneficial effects on neuromuscular symptoms and performance in glycogenosis type V (Vorgerd & Zange, 2007).

On the other hand, short- or long-term creatine supplementation did not influence cognitive processing in non-sleep deprived young adults (Rawson et al., 2008), functional capacity in peripheral arterial disease (Domingues et al., 2021), bone health in postmenopausal women with and without osteopenia (Lobo et al., 2015; Sales et al., 2020), muscle function in juvenile dermatomyositis (Solis et al., 2016), multiple sclerosis (Malin, Cotugna, & Fang, 2008) and tetraplegia (Kendall, Jacquemin, Frost, & Burns, 2005), patient- and clinician-reported outcomes in childhood-onset systemic lupus erythematosus (Hayashi et al., 2014) and amyotrophic lateral sclerosis (Drory & Gross, 2002; Shefner et al., 2004), symptoms of apnea of prematurity in infants (Bohnhorst et al., 2004), while showing negligible improvements in Parkinson's disease (Bender et al., 2006), patients with colorectal cancer (Norman et al., 2006), Huntington's disease (Bender et al., 2005), cystic fibrosis (Braegger et al., 2003) or rheumatoid arthritis (Willer, Stucki, Hoppeler, Brühlmann, & Krähenbühl, 2000).

Equivocal results of creatine trials in clinical medicine might be due to several factors, including the level of exposure to creatine and specific pathological conditions (un)responsive to creatine intervention. Creatine administration was typically short in duration (<3 months) and employed rather diverse dosage protocols (e.g., a daily amount administered varies from 0.25 g to over 20 g of creatine). A dose–response effect has been suggested in several studies (Vorgerd & Zange, 2007), with low-dose creatine often found as equivalent or even superior in terms of efficacy and safety to high dosages. This requires long-term and rigorously designed dose-escalation studies to find the optimal dose range for creatine. Creatine was dispensed either as a powder or

granules added to water solutions and foods, or solid dosage forms (e.g., tablets, capsules), with creatine monohydrate ($C_4H_{11}N_3O_3$) almost exclusively used as a chemical form of creatine. Still, monohydrate formulation might be relatively limited in terms of bioavailability, transport, and storage (Ostojic, 2020b), and alternative creatine mimetics have to be designed and critically scrutinized in clinical nutrition. Finally, the patients' characteristics could also affect the effectiveness of supplemental creatine, with a disease, nutritional status, and age recognized among the main factors. Creatine appears more beneficial in health conditions and diseases characterized by an impaired creatine synthesis/accretion, populations with a limited intake of creatine from the diet, and in older people, with the skeletal muscle-related symptoms and signs expected to improve substantially with the treatment. On the other hand, supplemental creatine was largely ineffective to tackle neurodegenerative diseases, bone health, and cancer. This may encourage further exploration of creatine in the general population and perhaps identify which subpopulations may profit from creatine and which are less sensitive.

5. Safety of creatine supplementation

During the past 30 years, creatine has been attentively investigated to detect, assess, understand, and prevent any side effects or adverse event linked to its use in human nutrition and medicine. A vast majority of pharmacovigilance studies demonstrated favorable safety of supplemental creatine, with creatine poses no adverse health risks in healthy people and clinical populations across various life stages and conditions, at dosages ranging from 0.03 to 0.8 g per kilogram of body weight per day for up to 5 years (for a detailed review see Balestrino & Adriano, 2019; Antonio et al., 2021). An initial mild weight gain (~1–2% of body mass) is the only consistently reported side effect from creatine supplementation (Eckerson, Bull, & Moore, 2008; Deminice et al., 2016; Almeida, Colombini, & Machado, 2020). This effect appears to be dose-dependent since lower doses of creatine (e.g., 0.03 g/kg/day) cause no weight gain or notable changes in body composition (Rawson, Stec, Frederickson, & Miles, 2011). In line with affirmative evidence from safety trials, the U.S. Food and Drug Administration (FDA) recently recognized creatine monohydrate as a safe ingredient (Generally Recognized as Safe, GRAS) (Food and Drug Administration, 2020), which labels creatine as a non-toxic food substance under the conditions of its intended use. A search through the Center for Food Safety and Applied Nutrition Adverse Event Reporting System (CAERS), a database that contains information on adverse event and product complaint reports submitted to the FDA for foods, dietary supplements, and cosmetics, revealed that creatine (and multi-component products containing creatine) was recorded in only 23 reports out of 30,766 (0.075%) (CFSA Adverse Event Reporting System, 2021). This further strengthens the proof of creatine harmlessness although any individual report requires attention and careful evaluation for systematic causality. Sporadic case reports described in the literature have been refuted in well-controlled clinical studies showing that creatine supplementation does not increase the incidence of gastrointestinal distress, musculo-skeletal injuries, or kidney dysfunction (for a detailed review, see Kreider et al., 2017). Interestingly, creatine appears safe when supplied through a regular diet at the population level. For instance, the odds ratio for having failing kidneys in U.S adults consuming ≥ 2.0 g/day of dietary creatine compared to low-intake counterparts (<1.0 g/day) was 0.74 (95% CI from 0.39 to 1.38) (Ostojic, 2021b), indicating no significant association between creatine intake and kidney dysfunction in the general public. Still, patients with renal impairment should not be treated with creatine unless careful analysis of the risk-benefit balance proves favorable (Balestrino & Adriano, 2019).

6. Should everybody take creatine?

The fact that creatine is exclusively available from meat-based foods

requires a prudent interpretation in respect of contemporary food policies and public health. Specifically, several intergovernmental organizations and research authorities have recently called for reorienting agricultural priorities by supporting less animal production and meat consumption in aim to tackle climate change, nurture human health, and support environmental sustainability (Jarmul, Liew, Haines, & Scheelbeek, 2019; Schiermeier, 2019; Willett et al., 2019). Although imaginably effective for mitigating and adapting to global nutritional challenges for the following decades, eating less meat could lead to an inadequate intake of several meat-specific nutrients, including creatine (Ostojic, 2020c). This perhaps allows considering forward-looking public strategies to supply enough creatine for optimizing its dietary load in food systems, including low-dose supplementation and food fortification.

Both strategies sound credible due to a relatively high prevalence of creatine deficiency, and the favorable safety and efficacy of creatine demonstrated in previous interventional studies. However, many issues remain to be addressed before recognizing the addition of creatine to food as a feasible public health intervention (Fig. 2). For instance, no dietary reference intakes for creatine are established since creatine is considered non-essential (Institute of Medicine, 2006), although the reports suggesting that creatine could be a conditionally essential nutrient begins to emerge (Reeds, 2000; Post, Tsikas & Bakker, 2018; Wu, 2020). Second, the optimal method for supplying creatine to the general public remains uncertain. Food fortification might be a convenient large-scale strategy to add creatine and improve the nutritional quality of the food supply. However, food fortification is usually carried out in front of severe symptoms of intake shortage in large shares of the population. For instance, a sustained salt iodization programs ensured an adequate iodine status and effectively controlled iodine deficiency across the globe (World Health Organization and Food Agriculture Organization, 2006). This has yet to be established for creatine, for which only epidemiological and preliminary data are provided at this moment. Still, as an odorless white powder, creatine could be efficiently added to foods commonly consumed by the general public (e.g., wheat flour, maize flour, cornmeal). Even so, we have to establish an appropriate level of fortification, and weigh production and retail costs to devise and offer affordable creatine-enriched foods to the general public. Opportunely, creatine can be produced industrially by several environmental-friendly processes, and the raw materials of these reactions (e.g., sarcosines, acetic acid, cyanamide, S-alkylisothiourea) are readily available as basic chemical components (Pischel & Gastner, 2007). Chemical synthesis of creatine requires no isolation from meat extracts yet whether creatine mass production is feasible on the scale needed to achieve food additive status remains undefined. In addition, creatine as creatine monohydrate may be unstable under certain conditions (e.g., low pH, high temperature, aqueous solutions), which requires scrupulous food processing and storage, and inventing more resilient alternatives. Food fortification may set off as a home fortification (e.g., adding creatine powder to convenient foods), followed by a volitional commercial and industrial fortification of foods for particular dietary purposes (e.g., meat-free products, or other foods designed for special dietary requirements from an individual), and mass fortification mandated and regulated by the government sector (Ostojic, 2020c). As a first step, creatine-fortified foods might be designed and pored over large populations of selected at-risk people, such as elderly people who may be frail and have an insufficient diet, vegans and vegetarians. This course of action (e.g., special voluntary fortification) between no general supplementation and universal supplementation might be well received and decrease the chances of this idea being unduly criticized.

The idea of providing creatine to the general population requires building an international network that should unite nutritional experts and regulatory authorities to build a case for creatine, perhaps by using a road map similar to choline, another compound that is endogenously produced and once was regarded as a non-essential compound but eventually recognized as an essential nutrient by the Food and Nutrition



Fig. 2. Advantages (green) and open questions (orange) for creatine use in the general population. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Board of the National Academies of Medicine ([National Academies of Medicine, 1998](#)). There is a critical need to collect more information about current dietary intakes of creatine across various age groups, and explore consequences of creatine deficiency and excess, followed by a selection of indicators for estimating the requirement for creatine and factors affecting these requirements. In addition, we have to better understand the public health significance of creatine malnutrition and the global burden of disease, evaluate the possible role of creatine supplementation in the control of creatine malnutrition, describe in details specific food vehicles for supplemental creatine, and implement sustainable supplementation/food fortification programs, at least in vulnerable population.

7. Conclusions

After over 100 years of scientific research about creatine, the time has come to perhaps reconsider this simple yet salient dietary compound as more than just another fancy supplement. The evidence shows that an inadequate intake of creatine from food could not be fully compensated by internal synthesis, suggesting creatine is an indispensable amino acid derivative for humans. Health consequences linked with low creatine intake found in ecological studies provide an additional framework for creatine value, and population-wide creatine malnutrition calls for scrutinizing strategies to provide creatine to the general public. Still, creatine has a long road ahead on the way to general acceptance as a food additive for all and sundry. Hopefully, this journey should start beyond sports science and clinical medicine, by encouraging public health authorities to think about creatine as a critical component of the

balanced low-meat diet of the XXI century.

Author contributions

S.M.O. drafted, edited, and revised the manuscript and approved the final version of the manuscript.

Ethical statement

This is a review paper, which doesn't include animal or human experiments.

Declaration of Competing Interest

S.M.O. serves as a member of the Scientific Advisory Board on creatine in health and medicine (AlzChem LLC). S.M.O. owns patent "Sports Supplements Based on Liquid Creatine" at European Patent Office (WO2019150323 A1), and active patent application "Synergistic Creatine" at UK Intellectual Property Office (GB2012773.4). S.M.O. has served as a speaker at Abbott Nutrition, a consultant of Allied Beverages Adriatic and IMLEK, and an advisory board member for the University of Novi Sad School of Medicine, and has received research funding related to creatine from the Serbian Ministry of Education, Science, and Technological Development, Provincial Secretariat for Higher Education and Scientific Research, AlzChem GmbH, KW Pfannenschmidt GmbH, ThermoLife International LLC, and Monster Energy Company. S.M.O. is an employee of the University of Novi Sad and does not own stocks and shares in any organization. The funders had no role in the design of the

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